

AGI Research Kit: Unified Preprint v1 [AGI Kit: An End-to-End Self-Improving Tool-Use Pipeline on Consumer Hardware](#) Authors: AGI Research Kit Contributors [Date](#): 2026-08-01 [Type](#): Single arXiv preprint, replacing the predecessor 5-paper TMLR bundle (archived).

Headline Numbers End-to-end task success: [68%](#) vs 30% static baseline (+38 pp) Continual-learning eval: [60.4% +/- 3.6%](#), [p<0.01](#) vs static A/B safety gate: [12/12](#) adversarial boundary tests passed Cross-model transfer: collapses below [~2B parameters](#) Hardware: ~3.3 GB RSS, ~20 s/episode, no GPU

Companion Artifacts Source: [src/agi_kit/](#), [experiments/](#), [scripts/](#) Logs: [logs/{cross_model,stat_tests,safety_gate,full_run*,continual,l4}](#) Real SFT validation: [data/sft_real/](#) (SmolLM2-135M, 2 min on CPU) Figures: [papers/figures/](#) (5 matplotlib PNGs at 200 DPI)

Why One Preprint Instead of Five? The five-paper TMLR bundle achieved an average reviewer score of 3.43 / 5.0 (Major Revision), driven by structural issues that additional polishing could not resolve: synthetic GAIA2 eval, no head-to-head baselines, small N. We honestly consolidated the bundle into a single preprint that frames the work as an empirical system report rather than a benchmark-beating contribution.
